

SIMPLICITY OF $\mathrm{PSL}_n(F)$

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1. SIMPLICITY OF $\mathrm{PSL}_n(\mathbb{F})$

For given field $\mathbb{F}$, the *projective special linear group* $\mathrm{PSL}_n(\mathbb{F})$ is defined by

$$\mathrm{PSL}_n(\mathbb{F}) = \mathrm{SL}_n(\mathbb{F})/Z(\mathrm{SL}_n(\mathbb{F}))$$

What is the center of $\mathrm{SL}_n(\mathbb{F})$? We claim that

$$Z(\mathrm{SL}_n(\mathbb{F})) = \{\lambda I_n \mid \lambda^n = 1 \in \mathbb{F}\}$$

Proof. First show that $h \in Z(\mathrm{SL}_n(\mathbb{F}))$ must be diagonal. If $h_{ij} \neq 0$ for $i \neq j$,

$$h \cdot \mathrm{diag}(1, \dots, x, \dots, x^{-1}, \dots, 1) \neq \mathrm{diag}(1, \dots, x, \dots, x^{-1}, \dots, 1) \cdot h$$

since (i, j) -th entry of left side is $x^{-1} \cdot h_{ij}$ but right side is $x \cdot h_{ij}$. Then $h \in Z(\mathrm{SL}_n(\mathbb{F}))$ is $\mathrm{diag}(h_1, \dots, h_n)$. For $I_n + E_{ij}$,

$$(I_n + E_{ij}) \cdot h = h \cdot (I_n + E_{ij})$$

it implies $h_i = h_j$ so $Z(\mathrm{SL}_n(\mathbb{F}))$ is set of scalar matrices in $\mathrm{SL}_n(\mathbb{F})$. □

Intuitively, we can view $\mathrm{SL}_n(\mathbb{F})$ acts on the set of 1-dim subspaces of vector space $\mathbb{F}^n$. In such perspective, the center $Z(\mathrm{SL}_n(\mathbb{F}))$ is the kernel of such action. It changes the ‘speed’ of 1-dim subspaces but not shapes. So,

$\mathrm{PSL}_n(\mathbb{F})$: acts on 1-dim line, change the angle of lines.

Definition 1.1. (Doubly Transitivity) G acts on X doubly transitive means that, for $x_1 \neq x_2$ and $y_1 \neq y_2$, there exists $g \in G$ such that

$$(x_1, x_2) \xrightarrow{g} (y_1, y_2)$$

equivalently, G acts transitively on X and let's fix $x \in X$. Then, $\mathrm{Stab}_G(x)$ acts on $X \setminus \{x\}$ transitively.

Proof. (Equivalence between 2 definitions) Suppose that first definition holds. Then, for $x' \in X \setminus \{x\}$ and any $y \in X \setminus \{x\}$, there exists $g \in G$ such that

$$(x, x') \xrightarrow{g} (x, y)$$

So $g \in \text{Stab}_G(x)$ and once we can choose y arbitrarily, $x' \mapsto y$ is transitive.

Suppose that second definition holds. For $x \neq x'$ and $y \neq y'$, we can find $y = g(x)$. Suppose that $g(x') = z$. Then we can find $h \in \text{Stab}_G(x)$ such that $h(z) = y'$. Let $k = g * h$. Then, $g * h(x) = g(x) = y$, and $g * h(x') = g(z) = y'$. $\square$

Theorem 1.2. $\text{PSL}_n(\mathbb{F})$ is simple $\forall n \geq 2$, $\forall \mathbb{F}$ a field except for $\text{PSL}_2(\mathbb{F}_2) \cong S_3$, $\text{PSL}_2(\mathbb{F}_3) \cong A_4$.

Our ultimate goal is prove this. To prove this, we'll construct bit story from now on. Actually the proof is strongly based on following criterion:

Theorem 1.3. (*Iwasawa's Criterion*) G acts doubly transitive on X .

- (1) There exists some $H_x = \text{Stab}_G(x)$ such that it contains $U \trianglelefteq H_x$, U is abelian, $\langle gUg^{-1} \mid g \in G \rangle = G$
- (2) G is perfect, i.e., $[G, G] = G$

Then, for the kernel K of G -action on X , G/K is simple.

Note that we already shown that $\text{SL}_n(\mathbb{F})$ acts doubly transitive on 1-dim vector spaces of $\mathbb{F}^n$ and kernel of such action is $Z(\text{SL}_n(\mathbb{F}))$. So, Iwasawa's criterion gives us that if $\text{SL}_n(\mathbb{F})$ holds (1) and (2), then $\text{PSL}_n(\mathbb{F}) = \text{SL}_n(\mathbb{F})/Z(\text{SL}_n(\mathbb{F}))$ is simple. (During the proof, exception cases occurs.) So first we prove following lemmata.

Lemma 1.4. G acts doubly transitively on X . For all $x \in X$, $\text{Stab}_G(x)$ is the maximal subgroup of G . i.e., no $K < G$ such that $\text{Stab}_G(x) < K < G$.

Proof. Choose arbitrary $x \in X$ and $H_x = \text{Stab}_G(x)$. We first claim that

$$G = H_x \cup H_x g H_x \text{ where } g \notin H_x \text{ and } H_x g H_x = \{hgh' \mid h, h' \in H_x\}$$

Let $g, g' \notin H_x$. Then, $g(x) \neq x$ and $g'(x) \neq x$. So using doubly transitivity,

$$(x, g(x)) \xrightarrow{h} (x, g'(x))$$

Since $h(x) = x$, $h \in H_x$ and so $(h * g)(x) = g'(x)$. However here, $(g')^{-1} h g(x) = x$ implies that $(g')^{-1} h g \in H_x$. So, $g' = hgh'$ for $h, h' \in H_x$. Since the choice was arbitrary, $G = H_x \cup H_x g H_x$.

Then suppose that $H_x \leq K \leq G$. For $k \in K \setminus H_x$, we can newly define that $G = H_x \cup H_x k H_x$. However, since $H_x \leq K$ and $k \in K$, it implies that $G \leq K$ so $K = G$. Since H_x is maximal. $\square$

Lemma 1.5. *Suppose that G acts on X doubly transitively, then $\forall N \trianglelefteq G$, N acts on X either trivially or transitively.*

Proof. Suppose that $N \trianglelefteq G$ acts on X non-trivially. i.e., $n(x) = x'$ where $x \neq x'$. Then for $(x, n(x))$ and any $y \neq y'$ in X , we can find $g \in G$ such that $g(x) = y$, $g(n(x)) = y'$. Hence,

$$(gng^{-1})(y) = y'$$

Since $y, y' \in X$ are chosen arbitrarily and $gng^{-1} \in N$, we can say that N acts on X transitively. $\square$

Now we're ready to prove Iwasawa's Criterion on doubly transitive G -action on X .

Proof. According to 4th Isomorphism theorem, the lattice structure of G/K is isomorphic to upper-structure of K in G -lattice. So, showing simplicity of G/K is equivalent to if $K \leq N \trianglelefteq G$ then $N = K$ or $N = G$. Suppose that

$$K \leq N \trianglelefteq G$$

And choose stabilizer of x , H from assumption (1), i.e., $U \trianglelefteq H$ and U is abelian, gUg^{-1} generates G . Since $N \trianglelefteq G$, we can know

$$H \leq NH \leq G$$

By lemma 1.4., H is the maximal subgroup. So, either $NH = H$ or $NH = G$. If $NH = H$, it implies that $N \leq H$ so N is normal subgroup such that stabilize x . By lemma 1.5., this implies that N acts on X trivially. Hence, $N \leq K$ and so $N = K$ in this case.

If $NH = G$, we claim that $NU \trianglelefteq NH = G$.

$$(nh)(n_1u)(nh)^{-1} = n_2(huh^{-1})n^{-1} = n_2u_2n^{-1}(u_2^{-1}u_2) = n_3u_2 \in NU \implies NU \trianglelefteq NH = G$$

However,

$$gUg^{-1} \leq gNUg^{-1} = NU \implies NU = G$$

By 2nd Isomorphism theorem,

$$G/N \cong U/(N \cap U)$$

However $U/(N \cap U)$ is abelian since U is abelian. By commutator subgroup theorem, G/N is abelian implies that $[G, G] = G \leq N$, so $N = G$. $\square$

Now what we want to show is actually $\mathrm{PSL}_n(\mathbb{F})$ is simple. To show that, it is enough to show that $\mathrm{SL}_n(\mathbb{F})$ satisfy Iwasawa's condition with kernel $Z(\mathrm{SL}_n(\mathbb{F}))$. We already show that $\mathrm{SL}_n(\mathbb{F})$ acts on 1-dim vector space of $\mathbb{F}^n$ and it is doubly transitive action, and the kernel is $Z(\mathrm{SL}_n(\mathbb{F}))$.

Lemma 1.6. $\mathrm{SL}_n(\mathbb{F})$ satisfies Iwasawa's 1st condition.

Proof. Choose H be the stabilizer of 1-dim vector space generated by standard basis $\hat{e}_1$. Then the block matrix representation of H will be

$$H = \left\{ \begin{pmatrix} a & * \\ 0 & [h] \end{pmatrix} \mid [h] \in \mathrm{GL}_n(\mathbb{F}), a = (\det h)^{-1} \right\}$$

We can find group homomorphism defined by

$$\varphi : \begin{pmatrix} a & * \\ 0 & [h] \end{pmatrix} \mapsto [h] \in \mathrm{GL}_n(\mathbb{F})$$

The kernel of this homomorphism $\varphi : H \rightarrow \mathrm{GL}_n(\mathbb{F})$ is easily given by

$$\ker \varphi = \begin{pmatrix} 1 & \vec{a} \\ 0 & I_n \end{pmatrix} \text{ where } \vec{a} = (a_1, \dots, a_{n-1})$$

Hence,

$$U = \ker \varphi \cong \mathbb{F}^{n-1} \implies U \trianglelefteq H$$

And via easy computation, U is abelian. To argue that such H and U holds Iwasawa's 1st condition, remaining to show is that gUg^{-1} generated $\mathrm{SL}_n(\mathbb{F})$. To show that, we'll first show that any unipotent elementary matrices ($I_n + \lambda E_{ij}$) conjugates to $I_n + E_{12} \in U$. Secondly, we'll show that unipotent elementary matrices generates $\mathrm{SL}_n(\mathbb{F})$. (So, gUg^{-1} contains all unipotent matrices and it generated $\mathrm{SL}_n(\mathbb{F})$.) Conjugate is peration basis changing. So, if we chance the basis for $T = I_n + \lambda E_{ij}$ from $\{e_i\}$ to $\{f_i\}$ such that $f_1 = \lambda e_i$, $f_2 = e_j$ and $f_3, \dots, f_n$ is permutation of remaining things. Then,

$$T : f_1 \mapsto T(\lambda e_i) = \lambda e_i = f_1$$

and

$$T : f_2 \mapsto T(e_j) = \lambda e_i + e_j = f_1 + f_2$$

and for remaining things, $T : f_k \mapsto f_k$. So the matrix representation of T in $\{f_i\}$ basis will be

$$T_f = \begin{pmatrix} 1 & 1 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ \vdots & & I_n & \end{pmatrix} = I_n + E_{12}$$

So, we can obtain

$$(I_n + \lambda E_{ij}) = g(I_n + E_{12})g^{-1}$$

for some $g \in \mathrm{GL}_n(\mathbb{F})$. To scaling g into $\mathrm{SL}_n(\mathbb{F})$, we just modify the basis $f_n = (\det g)^{-1}e_m$. It does not effect the result, but now $g \in \mathrm{SL}_n(\mathbb{F})$.

Now we need to show that set of unipotent elementary matrices generates $\mathrm{SL}_n(\mathbb{F})$.

□

Lemma 1.7. $\mathrm{SL}_n(\mathbb{F})$ satisfies Iwasawa's 2nd condition.

Proof. Sufficient to show that all unipotent elementary matrices are commutator. Since commutators are conjugated to each other, sufficient to show that $I_n + E_{12}$ is a commutator. Do $[I_n + E_{13}, I_n + E_{32}]$. Here exception occurs: for $n = 2$,

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}^{-1} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & b(a^2 - 1) \\ 0 & 1 \end{pmatrix}$$

So, $\mathbb{F}_2, \mathbb{F}_3$ is impossible.

□

Hence, $\mathrm{PSL}_n(\mathbb{F}) = \mathrm{SL}_n(\mathbb{F})/Z(\mathrm{SL}_n(\mathbb{F}))$ is simple except for $\mathrm{PSL}_2(\mathbb{F}_2) \cong S_3$ and $\mathrm{PSL}_2(\mathbb{F}_3) \cong A_4$.